



Global optimization

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The current status of global optimization in computational molecular science is characterized in this article by focusing on one particular area, evolutionary algorithms applied to cluster structure optimization. Other algorithms and application areas are also mentioned briefly, indicating a widespread use of global optimization techniques in every conceivable branch of computational molecular science. © 2011 John Wiley & Sons, Ltd. *WIREs Comput Mol Sci* 2011 1 879–887 DOI: 10.1002/wcms.70

INTRODUCTION

Optimization of a function is the task of finding one of its extrema. Clearly, maximization can be turned into minimization, and vice versa, by a simple overall sign change of the given function. Therefore, minimization (or maximization) and optimization are often used rather indiscriminately. For convex functions, finding a minimum is tantamount to finding the lowest possible function value, and this value can be found with a single local minimization (essentially, by stepping only downhill) from every starting point. In many interesting applications, however, one faces a nonconvex function, with many minima of different function values. Then, finding just one minimum tells us little or nothing about the globally minimal function value. Also, which local minimum we find will depend strongly on which starting point we happened to choose.

In contrast, a truly global minimization should be able to find the globally minimal function value, irrespective of the starting point. Clearly, in a nonconvex case, such a deterministic global optimization yields very valuable information. In practice, however, it turns out that deterministic global minimization often is extremely expensive, to the point of infeasibility. Fortunately, stochastic algorithms using certain problem-solving heuristics can frequently be employed to quickly find minima with function values not too far above the global minimum in a small fraction of the time needed for a deterministic search for the true global minimum. Therefore, global optimization methods have found increasing use in many areas of chemistry in the recent decades. This paper

gives an impression of this current usage of global optimization methods, by focusing on one broad application area, global cluster structure optimization, and one method, evolutionary algorithms, but also giving examples of applications of different algorithms in very different areas. With the given space limitations, no exhaustive review is possible in any of these cases, instead the given references merely indicate typical examples of recent work.

DETERMINISTIC VERSUS STOCHASTIC OPTIMIZATION

Optimization is a broad field of mathematics, with several different subfields specialized on problems with given characteristics that can be exploited for efficient solution. Comprehensive overviews on recent developments¹ and future trends² have been provided in recent years. Here, the focus is on one of these fields, nonlinear, nonconvex global optimization with little or no exploitable prior information.

Several reviews have covered methods³ and applications⁴ of deterministic global optimization, documenting the advanced development stage and the widespread practical use of such techniques. Very briefly, to be sure of having found the global optimum of a function, one must have visited all search space. For continuous variables, and without further ‘tricks’, this would entail an infinite number of function evaluations already in a one-dimensional (1D) case. Hence, global optimization is all about finding clever ways to ‘judge’ finite patches of search space without visiting all points in them. One important way of doing that is the ‘branch and bound’ technique. It amounts to separating search space in subregions (‘branches’) and then to employ cleverly chosen (under- or over-)estimators (‘bounds’) to decide if a given region might contain a better point than the best one found so far, in which case this subregion

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is subdivided further. Recently, a comparison⁵ was made of deterministic global (constrained) optimization methods that perform a complete search of the whole search region, on a test suite of over 1000 problems in up to 1000 variables. As to be expected, the performance of all methods degrades strongly with dimensionality; within a preset time of half an hour the best method (BARON) solves 90% of the problems with less than 100 variables, but only 63% of the larger ones. Despite claims of exactness, most methods occasionally come up with wrong findings.

Sadly, however, the global search of cluster potential energy surfaces turns out to be a particularly hard global optimization problem. In addition, deterministic global search depends on prior knowledge of the interparticle forces. Correspondingly, an early attempt⁶ at deterministic global cluster structure optimization was limited to Lennard–Jones (LJ) clusters of only seven atoms. Several years later, an extension to molecular clusters was published,⁷ tailored to the common optimized potentials for liquid simulations (OPLS) force-field, but the largest system considered consisted of one single *n*-butane next to one single ethylamine molecule. Even more recently, deterministic global searches of molecular conformations were still limited to rather small systems, as exemplified by the work of Lin and Stadtherr⁸ who employed interval arithmetic methods to find optimal conformations of alkanes. Using realistic empirical potentials, they were limited to undecane in a united-atom model, i.e., to a problem with only 11 explicit united-atom centers. Similarly, using methods of the branch-and-bound type, Beliakov and Lim⁹ could optimize configurations of small oligopeptides with 10–14 explicitly treated dihedrals, but failed for cases with 24 dihedrals, mainly because of the exponentially increasing memory need for storing lower bounds on partitioned configurational subspaces. Again, this represents little progress in terms of problem size because the very same test molecules (Met-Enkephalin and variants of it) had already been successfully treated with α -branch-and-bound methods several years earlier,¹⁰ even including an approximation to entropic contributions for optimizing free energies.

In contrast, stochastic–heuristic global optimization methods have seen several decades of applications to significantly larger problems, at the expense of forgoing any certainty of the results or even of the convergence behavior. The theoretical and practical differences between these two algorithm classes are much larger than any differences within these two classes. In fact, comparisons between stochastic global optimization methods are fairly rare and rather controversial because many of them can be

tuned to a given problem to varying extents and this has a strong impact on performance. As an example, in a comparison¹¹ of basin hopping with evolutionary algorithms, the latter was found to be superior to the former except for cases where further algorithmic tuning was required. Nevertheless, the same outcome could also be interpreted as basin hopping being easier to apply in diverse areas because it needs less application-specific tuning. (Evolutionary algorithms are explained below, in the subsection ‘refined technical details’. Basin hopping essentially combines Monte-Carlo-like random moves with local optimization.)

TECHNIQUES AND APPLICATIONS OF STOCHASTIC GLOBAL OPTIMIZATION

This section delineates fundamental and advanced issues in designing efficient stochastic global optimization algorithms for practical use, frequently interspersed with examples of their application.

The Basic Ingredients

Successful stochastic–heuristic global optimization methods have to excel in one or more of the following basic aspects:

- Find a local minimum quickly, from a given starting point.
- Get out of this minimum again quickly.
- Do not get lost in irrelevant detail but instead focus upon the overall structure of the function to be optimized.
- Avoid both complete enumeration of all minima (which would elevate the method to a deterministic one, with all the associated expense) and blindly jumping into unexplored regions but instead direct jumps to promising regions by exploiting information accumulated in previous stages of the global optimization run.

Interestingly, it has been shown that each of these ingredients on its own can be turned into a global optimization scheme that is successfully applicable in practice. Item (a) has been employed in the form of series of local optimizations long before the advent of global optimization as such, with sets of starting points generated by any means (by chemical intuition, at random, or by sampling from a high-temperature trajectory, etc). Given the advanced

development state of local optimization methods, in particular, as employed for extremely expensive *ab initio* quantum chemistry tasks, it is hard to improve upon them further. The choice of starting points, however, can be improved beyond the simple ones just mentioned. It turns out that ‘hybridizing’ global with local search (i.e., performing a local search on each trial solution a global search step delivers) is a significant improvement, compared to both nonhybridized global search and raw series of local searches. This has been attributed¹² to local optimization transforming the potential energy surface into a staircase structure, which eliminates many barriers and emphasizes the low-resolution funnel structure (item c).

Item (b) is trivial to realize in methods that take big jumps across search space anyway, but is of importance for other methods that progress in smaller steps, such as MD-based simulated annealing, small-step tabu-search,¹³ or basin-hopping.¹⁴ Improving upon this point is central to recent modifications¹⁵ of the basin-hopping strategy and to the proposed use of time-reversed¹⁶ dissipative dynamics.

Besides the staircase transformation, another way to achieve item (c) is to use a quick method for scanning the neighborhood of a recently found minimum. There have been multiple attempts in this direction, from the ‘directed mutation’ by Hartke¹⁷ to the ‘funnel hopping’ scheme by Cheng et al.¹⁸

Item (d) is the premise of all population-based methods, including not only evolutionary algorithms (EAs) but also the broad variety of methods building more explicitly upon ‘communication’ between many trial solutions [swarm intelligence, ant-colony, etc., but also covariance matrix adaption evolution strategy (CMA-ES) and others]. Again, the items are not really independent of each other: Adding several aspects of population-based global search methods to a greedy single-individual search was found¹⁹ to greatly improve escape from distracting funnels.

Refined Technical Details

In contrast to the previous, very general subsection, the present subsection focuses strongly but not exclusively on EAs [or genetic algorithms (GAs) as a particular realization of them] to the specific problem of global cluster structure optimization. Fundamental introductions to EAs/GAs can be found in textbooks²⁰; for completeness, a short sketch is given in the following paragraph. A brief review of the whole theoretical toolbox needed for studying clusters (methods for energy calculation, global minimization techniques, and dynamics) was provided recently,²¹ together with their application to ZnO, ZnS, SiO₂,

and TiO₂ clusters. Another review²² more specifically focused on how to design genetic algorithms for this task. Complete monographs²³ have already been devoted to EA/GA applications in chemistry. A compact version of the personal view of the present author on the past development and future challenges of the field have been given in the introduction of a recent paper.²⁴ Therefore, the present remarks attempt to go beyond those stages.

Very briefly, EA procedures typically start with setting up a ‘population’ of ‘individuals’, each of them representing a point in search space. These are either chosen at random or contain some prior knowledge, or both. Evaluating the objective function for these points generates a first set of trial solutions to the problem. If the EA is hybridized with local search, each of these individuals is relaxed into a local minimum before proceeding further. Then, new trial solutions are generated from the previous ones, by a ‘mating’ procedure: Two ‘parents’ are chosen, with some preference for those that achieved the best objective function values. The information from these two parents is mixed (‘crossover’), generating one or two new trial points (‘children’). With a given probability, each child is changed (‘mutated’) in a random fashion, and again relaxed to a local minimum in case of a hybrid EA. Children with better objective function values replace parents, either periodically (generational EA) or continuously (steady-state or pool EA), keeping population size constant. From a practical point of view, the main difference between a GA and a (phenotype) EA is that in the former each point in search space is represented as a 1D string of (decimal or binary) numbers²⁵ on which crossover and mutation are then performed in an abstract fashion. In contrast, in the latter some physical meaning of the search space point is retained, i.e., in the case of cluster structure optimization it simply is one particular cluster conformation in three-dimensional (3D) space, and operators such as crossover and mutation are then performed directly²⁶ on this 3D representation. Randomness enters in many places, e.g., in the setup of the initial population, in the choice of parents for reproduction, in the crossover and mutation operators, and in replacement of individuals. Nevertheless, heuristic elements are also present, e.g., in the design of the crossover and mutation operators. It seems reasonable to assume that the whole procedure works because of a suitable balance between items (a) and (b) (cf. above), and because item (d) seems to be realized via crossover: In cluster structure optimization, a typical phenotype crossover cuts each parent cluster in halves and reassembles them in a crossover fashion to generate two children. Clearly, if a low-energy cluster

structure is cut in two parts, each of the parts is also still low in energy. Likewise, if low-energy parts are reassembled in a sensible manner, avoiding both too short distances (clashes) and too long ones (dissociation), there is a good chance that the resulting whole cluster will also be of low energy. This is mere speculation, however. Embarrassingly, there is an acute lack of proofs in this area. There is Holland's schema theorem²⁷ stating (roughly adapted to the problem of cluster structure optimization) that the probability to encounter small, low-energy cluster parts is increasing exponentially as the EA/GA progresses. Beyond that, few hard facts are known, and thus most EA practitioners resort to believing mostly in their own accumulated practical experience—which states that EAs work very well in practice. Therefore, algorithmic refinements of them have reached a rather advanced stage by now, as illustrated in the following.

The 'directed mutation' operator mentioned above is a chimera between a deterministic and a stochastic procedure. It takes the particles contributing least to the overall binding energy of the cluster (which is trivial to compute for typical force fields or other energy expressions based on a few-body expansion of the total energy) and moves them to 'holes' where they contribute most. This idea was extended to the so-called interior/exterior operators,²⁸ which were shown to be able to globally optimize cluster structures on their own. This was feasible not only for atomic clusters but also for molecular ones, including orientation moves,²⁹ in an application to benzene clusters. New global minima were found for $n = 11, 14, 15$; first proposals for global minima were made for $n = 16–30$.

For water clusters, certain local patterns of hydrogen bond arrangements are known to be better than others. Therefore, it is possible to boost the search for optimal water clusters (without unduly biasing it) by introducing mutation moves that modify these h-bond arrangements in a controlled way, without altering the center-of-mass positions of the water molecules. This was shown to be successful^{30,31} in two different algorithmic embeddings.

In many applications, the ranges in which parameter or coordinate values are allowed to be varied within a GA have to be preset in advance. Adjusting these bounds to what the EA itself discovers as the most likely values, so-called 'cultural evolution', speeds up convergence of evolutionary algorithms, as shown for crystal structure solution³² from powder diffraction data. Dissimilarity measures help in preventing collapse of a search population into a few easily accessible minima, as demonstrated in an application³³ of the population basin hopping method

to Morse clusters. Contrary to conventional recommendations, a theta function-like probability distribution has been proven to be optimal³⁴ for selecting individuals as parents in the crossover operation.

Greedy search of which lattice positions should be occupied in a predefined lattice is a discrete global optimization problem and hence much faster than continuous search in cluster configuration space. This has been used in the Dynamic Lattice Search method^{35,36} and its applications to LJ clusters and similar systems. Of course, it then is of paramount importance not to bias the lattice setup too strongly toward one single structural type. Introducing different lattice types³⁷ can easily lead to lower-energy clusters.

Operating global cluster structure optimization directly at the DFT³⁸ or *ab initio*³⁹ level has been attempted several times but was always found to be too expensive to be applicable to clusters of a size for which global optimization is obviously necessary, while retaining global convergence to some practically useful degree. The typical remedy in such situations is the two-stage procedure of performing the global search at some cheap force-field or semiempirical⁴⁰ level, followed by a DFT or *ab initio* refinement of the best candidates from the first stage. Obviously, the reliability of the overall procedure hinges upon a sufficiently good correlation of the structure-energy ordering between the two theory levels; ideally, this should be tested.⁴¹ The correlation can be improved by iterating both steps in a closed-loop fashion,⁴² alternating between structural optimization and optimization of low-level-theory parameters.

How to modify GAs in the presence of noise⁴³ is another topic of ongoing research. So far it has not been incorporated into theoretical chemistry applications such as cluster structure optimization or conformational search; it could conceivably be used, however, in the context of coarse-grained models of cluster/molecule environments. For self-assembly of more complicated particles (heterogeneous molecular clusters) it has been found⁴⁴ that Monte-Carlo moves are outperformed by rather simple heuristical moves such as building optimal larger clusters from optimal smaller ones.

Clearly, with two or more types of particles in a cluster, the complexity of the global optimization problem increases markedly, due to the combinatorial explosion of possibilities for type exchange on given positions in 3D space. Therefore, studies of binary clusters are far less numerous than that of homogeneous ones. All possible compositions of a binary variant of the standard LJ system⁴⁵ were studied for several exemplary cluster sizes, and

several composition-induced structural transitions were found. Binary metallic clusters have been examined by the Johnston group,⁴⁶ emphasizing the benefits of introducing a special type-exchange operator and focusing on size- and species-dependent propensities for segregation or mixing. Sokol et al.⁴⁷ showed that using EAs to generate a pool of diverse cluster structures for a few representative binary systems allows for reliable predictions of cluster structures for several other, related binary systems. Binary clusters of simple ionic species such as NaCl also have received some attention, both with empirical potentials⁴⁸ and with *ab initio* methods.⁴⁹ Given the combinatorial explosion mentioned above, it is hardly surprising that studies of ternary clusters⁵⁰ are quite rare, and that of higher-order clusters⁵¹ have only just begun.

For homogeneous clusters, it is still an issue how and at which cluster size⁵² structural patterns similar to the bulk system emerge. Finding the lowest energy cluster structures, however, is only a first step. In combination with statistical thermodynamics, the actual equilibrium distributions of cluster sizes and structures can be approximately predicted. For water clusters,⁵³ this substantially modifies the picture based only on potential energy. As a case rather similar to cluster structure optimization, not surprisingly GAs have also been employed to find optimal surface reconstruction⁵⁴ patterns. By now, this is also an application area large enough to require specialist reviews.⁵⁵ Obviously, it is also possible to use basically the same global optimization techniques on clusters not at the atomic/molecular level but at lower resolution. This was recently exemplified by studies^{56,57} on self-assembly of model particles for virus capsids and micelles. Only very recently, it was realized that global cluster structure optimization can also be inverted, to generate optimal vacancy clusters⁵⁸ in bulk lattices.

Global structure prediction can actually become simpler if additional constraints are imposed. This is exemplified by crystal structure prediction,⁵⁹ where frequently complete enumeration techniques generating a representative approximation to covering all search space are employed, combined with a sophisticated mix of models for the different components of the interaction energy: With such techniques, successful predictions of crystal structures for fairly complicated rigid molecules⁶⁰ or moderately complicated flexible molecules⁶¹ are well within reach. Tough cases of mixtures of flexible organic molecules with packings differing in intramolecular conformations and intermolecular hydrogen bonding patterns can also be solved.⁶² Five community-wide blind tests⁶³ of prediction software have been run. The related

BOX 1: THE SEARCH 'LANDSCAPE'

For a long time, more intuitively understandable connections between the macroscopic dynamical behavior of a system and the forces acting between its constituent particles were sought. Some illumination could be provided by inserting the 'landscape' concept^{65–67} as an explanatory bridge between them. As one specific aspect of this sort, the dependence of cluster structures on the range of a pairwise potential has been studied frequently, recently even for rather large clusters.⁶⁸ Clearly, stochastic-heuristic global optimization methods are indispensable tools for exploring and navigating this landscape. They have become so effective that it is a challenge to design landscapes of controlled difficulty⁶⁹ and known global optimum suitable for testing them.

problem of determining solid-state molecular structure from powder X-ray diffraction data has also been treated using GAs and suitably pre-defined⁶⁴ molecular fragments.

Further Application Areas

This subsection serves to dispel a possible misinterpretation of the previous subsection, namely that applications of global optimization in chemistry have been limited to EA/GA treatments of cluster structure optimization and to very similar topics. There are very many, very diverse further application areas, and many of them could be discussed in a similarly detailed manner. Instead, only a few introductory links to a few application areas are given here.

Protein folding⁷⁰ clearly is yet another global optimization problem, similar to cluster structure optimization but with the additional constraint that the amino acid sequence (primary structure) is given and should not be altered in the optimization process. Applications of GAs in this area range from GA design considerations⁷¹ via free-energy landscape studies⁷² (cf. Box 1) to the actual folding of whole proteins⁷³ themselves. There are indications⁷⁴ that the actual folding itself is not the most difficult part of the problem. Instead, the best folds found depend qualitatively on the force field used; therefore designing better protein force fields⁷⁵ would greatly improve *ab initio* protein folding predictions (without recourse to known folds from protein databases). Reality may even be more complicated than that because other factors such as glycosylation⁷⁶ may also influence protein folding *in vivo*. Conformational search for flexible molecules can be considered as a related subproblem, as exemplified by the usage of an ant-colony

algorithm⁷⁷ to find optimal conformations for a range of molecules, from alkanes via drug-like molecules to vitamin A.

Deducing molecular structure from experimental atomic distance data (usually obtained from NMR techniques) is called the molecular distance geometry problem. It has advanced to a stage beyond practical applications, as a mathematical problem in itself, as discussed by Liberti et al.⁷⁸ Interestingly, if all pairwise distances are known and exact, the problem can be solved in linear time.⁷⁹ In chemical applications, however, neither is the set of distances complete nor are the distances known exactly. Then the problem is strongly NP-hard⁸⁰ (implying exponential size scaling in practice) and is usually tackled with stochastic-heuristic global optimization approaches.⁸¹

Molecular docking (usually with the aim to find new inhibitor molecules blocking active sites in proteins) is yet another global optimization problem that has grown into a whole world of its own, as described in a brief but informative review.⁸² Likewise, GA applications in materials science abound, as reviewed⁸³ recently.

GAs have also been used extensively in laser pulse design for controlling photochemical reactions, both in experiments^{84,85} and in theory.⁸⁶ For the same purpose, also other stochastic global optimization methods have been used, for example the ant-colony algorithm.⁸⁷ The whole area of optimal control theory in chemistry and physics has been reviewed⁸⁸ recently.

In general, stochastic global optimization methods have been applied in most areas where the need for global optimization is obvious. For example, GAs have been used for a long time in (re)optimizing parameters of force fields⁸⁹ or semiempirical⁹⁰ methods, relative to *ab initio* reference data, or in optimizing

basis sets⁹¹ for quantum chemistry calculations. Astonishingly, however, global optimization methods of the same kind have also been used in far more exotic settings, for example to solve the Schrödinger equation⁹² or to determine placement and form of nodal surfaces⁹³ in diffusion quantum Monte-Carlo calculations.

The ultimate dream of chemistry is to design optimal molecules for performing particular functions. This task has already been tackled with global optimization methods, for example for photochemical properties of molecules,⁹⁴ catalytic transition metal complexes,⁹⁵ conducting polymers,⁹⁶ and design concepts for artificial photochemical energy conversion.⁹⁷

CONCLUSION

In the last 25 years, global optimization in theoretical molecular science has progressed from a rare curiosity in few selected areas to a well-established set of methods with proven applicability or even routine usage almost anywhere. The final transition into an everyday commodity for the nonexpert has not quite happened yet, as witnessed by the absence of global optimization capabilities from most quantum chemistry and molecular dynamics program packages, in stark contrast to the ubiquity of efficient local optimization. Clearly, to ease this transition, it is necessary to generalize existing specialized implementations. First steps⁹⁸ in this direction, however, have already been taken. It is easy to predict that stochastic-heuristic methods will take the lead in the first wave. It remains to be seen how quickly deterministic methods will be able to catch up.

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